COHERENT SHEAVES ON THE NOETHERIAN SPACE ARE QUOTIENTS OF SUMS OF PUSHFORWARDS OF POWERS OF THE IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 13452-13535, immediately following the statement at 13439-13450.

Editorial note. The displayed chain identifying global sections is retained verbatim; it is the technical heart of the proof.

Editorial note. Five cross-chapter pointers (Etale Cohomology, Coherent, Descent on Spaces, Morphisms of Spaces) are recorded in assembly.external_references.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 13519 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- The situation of descending the resolution property. chapter situation situation-descend-resolution-property, cited in the statement. Fixes the finite locally free covering pi : Y -> X and the classifying ideal I.
- Etale splitting off of a single quasi-finite part. chapter lemma lemma-etale-splits-off-just-one-quasi-finite-part, cited in the proof. Produces the etale neighbourhood and the single-point finite piece used to construct local generators.
- Duality of finite modules over the covering. Derived Categories of Spaces lemmas on finite modules over open pieces and Spaced cohomology lemmas, cited in the proof. Identifies the module of homomorphisms with a global section module so that generators can be transported.
- The canonical bundle of the finite locally free covering. the sheaf omega_{Y/X} appearing in the proof (dualising module of the covering). Turns sections of pi^*F twisted by omega into sections of F, which is how the displayed sum of pushforwards is produced.

Source reference. Original reference 04HL: focused-seeds/geometry/S9-spaces-surjection-noetherian-case/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0ADU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7544–7544. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 0GUY: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13416–13416. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0DK4: focused-seeds/geometry/S9-spaces-surjection-noetherian-case/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07UM: focused-seeds/geometry/S9-spaces-surjection-noetherian-case/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 060Z: focused-seeds/geometry/S9-spaces-surjection-noetherian-case/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08AI: focused-seeds/geometry/S9-spaces-surjection-noetherian-case/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 0GUY, assume X is Noetherian. Then for any coherent $\mathcal{O}_X$-module $\mathcal{F}$ there exist $r \geq 0$, integers $n_1, \ldots, n_r \geq 0$, and a surjection*

$$\bigoplus\nolimits_{i=1,\ldots,r} \pi_*(\mathcal{I}^{n_i}) \longrightarrow \mathcal{F}$$

of $\mathcal{O}_X$-modules.

Proof. Denote $\omega_{Y/X}$ the coherent $\mathcal{O}_Y$-module such that there is an isomorphism

$$\pi_*\omega_{Y/X} \cong \mathcal{H}om_{\mathcal{O}_X}(\pi_*\mathcal{O}_Y, \mathcal{O}_X)$$

of $\pi_*\mathcal{O}_Y$-modules, see Morphisms of Spaces, Lemma 08AI and Descent on Spaces, Lemma 060Z. The canonical map $\mathcal{O}_X \to \pi_*\mathcal{O}_Y$ produces a canonical map

$$\mathrm{Tr}_\pi : \pi_*\omega_{Y/X} \longrightarrow \mathcal{O}_X$$

Since V is Noetherian affine we may choose sections

$$s_1, \ldots, s_r \in \Gamma(V, \pi^*\mathcal{F} \otimes_{\mathcal{O}_Y} \omega_{Y/X})$$

generating the coherent module $\pi^*\mathcal{F} \otimes_{\mathcal{O}_X} \omega_{Y/X}$ over V. By Cohomology of Spaces, Lemma 07UM we can choose integers $n_i \geq 0$ such that s_i extends to a map $s_i' : \mathcal{I}^{n_i} \to \pi^*\mathcal{F} \otimes_{\mathcal{O}_Y} \omega_{Y/X}$. Pushing to X we obtain maps

$$\sigma_i : \pi_*\mathcal{I}^{n_i} \xrightarrow{\pi_* s_i'} \pi_*(\pi^*\mathcal{F} \otimes_{\mathcal{O}_Y} \omega_{Y/X}) = \mathcal{F} \otimes_{\mathcal{O}_X} \pi_*\omega_{Y/X} \xrightarrow{\mathrm{Tr}_\pi} \mathcal{F}$$

where the equality sign is Cohomology of Spaces, Lemma 0DK4. To finish the proof we will show that the sum of these maps is surjective.

Let $x \in |X|$ be a point of X. Let $v \in |V|$ be a point mapping to x. We may choose an étale neighbourhood $(U, u) \to (X, x)$ such that

$$U \times_X Y = W \coprod W'$$

(disjoint union of algebraic spaces) such that $W \to U$ is an isomorphism and such that the unique point $w \in W$ lying over u maps to v in $V \subset Y$. To see this is true use Lemma 0ADU and Étale Morphisms, Lemma 04HL. After shrinking U further if necessary we may assume W maps into $V \subset Y$ by the projection. Since the formation of $\omega_{Y/X}$ commutes with étale localization we see that

$$\pi_*\omega_{Y/X}|_U = (\pi|_W)_*\omega_{W/U} \oplus (\pi|_{W'})_*\omega_{W'/U}$$

We have $(\pi|_W)_*\omega_{W/U} = \mathcal{O}_U$ and this isomorphism is given by the trace map $\mathrm{Tr}_\pi|_U$ restricted to the first summand in the decomposition above. Since W maps into V we see that $\mathcal{I}^{n_i}|_W = \mathcal{O}_W$. Hence

$$\pi_*(\mathcal{I}^{n_i})|_U = \mathcal{O}_U \oplus (W' \to U)_*(\mathcal{I}^{n_i}|_{W'})$$

Chasing diagrams the reader sees (details omitted) that $\sigma_i|_U$ on the summand $\mathcal{O}_U$ is the map $\mathcal{O}_U \to \mathcal{F}$ corresponding to the section

$$s_i|_W \in \Gamma(W, \pi^*\mathcal{F} \otimes_{\mathcal{O}_Y} \omega_{Y/X}) = \Gamma(W, \mathcal{F}|_W \otimes_{\mathcal{O}_W} \omega_{W/U}) = \Gamma(U, \mathcal{F})$$

Since the sections s_i generate the module $\pi^*\mathcal{F} \otimes_{\mathcal{O}_Y} \omega_{Y/X}$ over V and since W maps into V we conclude that the restriction of $\bigoplus \sigma_i$ to U is surjective. Since x was an arbitrary point the proof is complete. □